

Why Quantum Mechanics?

Classical Physics Was Beautiful, But Nature Had Other Plans

Quantum Chemistry Notes

Nimeesh

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Big Idea

Quantum mechanics was not created because physicists wanted nature to sound mysterious. It was created because classical physics started failing when people looked carefully at light, atoms, and radiation.

1 Why even ask this question?

Before jumping into wavefunctions, operators, Hamiltonians, orbitals, and all the scary-looking quantum chemistry machinery, I think it is worth asking a simple question first:

Question

Why was quantum mechanics needed in the first place?

Classical physics was not some weak theory. It was extremely successful. Newton's laws explained the motion of planets, falling bodies, pendulums, and machines. Maxwell's equations explained electricity, magnetism, and light.

For a long time, it genuinely looked like physics was almost complete.

The classical idea was basically this:

If we know the position, velocity, and forces acting on a system, we can predict its future.

This works beautifully for large objects. But when physicists started studying atoms, light, radiation, and electrons, this classical picture began to crack.

The three major problems I want to focus on are:

1. Blackbody radiation
2. Photoelectric effect
3. The atom problem: atomic stability and line spectra

These were not small errors. These were deep failures.

2 Problem 1: Blackbody Radiation

2.1 What is a blackbody?

A perfect blackbody is an ideal object that absorbs all electromagnetic radiation falling on it. It does not reflect or transmit any radiation.

When it is heated, it emits radiation.

At room temperature, most of this radiation lies in the infrared region, so we cannot see it with our eyes. But with an infrared camera, we can detect it.

As the temperature increases, the emitted radiation shifts toward the visible region. That is why a hot object first glows red, then yellowish, and eventually white-hot.

Note to Self

This already tells me something important: temperature affects the distribution of emitted wavelengths.

2.2 The classical prediction

Classical physics tried to explain the spectrum of radiation emitted by a blackbody using the Rayleigh-Jeans law.

For spectral radiance as a function of frequency, the classical result is:

$$B_\nu(T) = \frac{2\nu^2}{c^2} k_B T$$

where:

- $B_\nu(T)$ is the spectral radiance,
- ν is frequency,
- c is the speed of light,
- k_B is Boltzmann's constant,
- T is temperature.

The problem is that this formula predicts that as frequency increases, the emitted radiation keeps increasing without limit.

So at high frequencies, especially in the ultraviolet region, classical physics predicted an absurd result:

A blackbody should emit an infinite amount of energy.

This was called the **ultraviolet catastrophe**.

But real blackbodies do not behave like this. Experimentally, the curve rises, reaches a peak, and then falls.

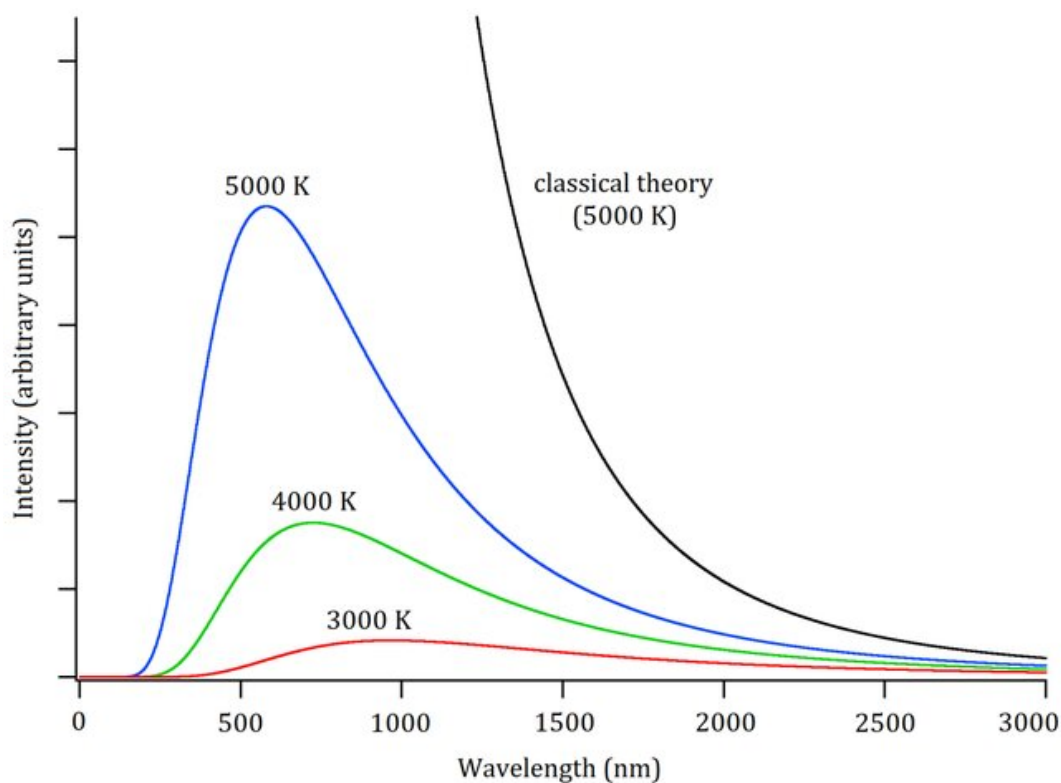


Figure 1: Comparison between the observed blackbody spectrum and the classical Rayleigh-Jeans prediction, which diverges at short wavelengths. Figure uploaded by David S. Bradshaw.

2.3 Planck's idea

Planck solved this by making a very bold move.

He said energy is not exchanged continuously. Instead, energy is exchanged in small packets called **quanta**.

The energy of one quantum is:

$$E = h\nu$$

where:

- E is the energy,
- h is Planck's constant,
- ν is frequency.

This means high-frequency radiation needs larger energy packets.

The average energy of a mode becomes:

$$\langle E \rangle = \frac{h\nu}{e^{h\nu/k_B T} - 1}$$

This equation is the real killer.

At low frequencies, $h\nu$ is small compared to $k_B T$, so the result behaves almost classically.

But at high frequencies:

$$h\nu \gg k_B T$$

The energy packet becomes too expensive compared to the available thermal energy. So those high-frequency modes are rarely excited.

The exponential term:

$$e^{h\nu/k_B T}$$

grows very fast, which makes the average energy fall sharply.

So the ultraviolet part does not blow up to infinity anymore. It drops.

Big Idea

Planck solved the blackbody problem by changing one basic assumption: energy is not exchanged smoothly at the microscopic level. It comes in packets.

3 Problem 2: Photoelectric Effect

3.1 The classical expectation

Maxwell's theory treated light as a wave.

In a wave picture, increasing the intensity of light means increasing the amplitude of the wave. More intensity means more energy.

So classical physics expected this:

If we shine brighter light on a metal, electrons should eventually be emitted.

This sounds reasonable. If we give more energy to the electrons, they should escape more easily.

But experiments showed something else.

3.2 What actually happened

For some frequencies of light, no electrons were emitted at all, no matter how intense the light was.

But if the frequency was high enough, electrons were emitted immediately.

So the key factor was not just intensity. It was frequency.

This was strange because classical wave theory had no clean explanation for why frequency should decide whether electrons are emitted.

3.3 Einstein's explanation

Einstein used Planck's idea and said that light itself comes in packets called **photons**.

Each photon has energy:

$$E = h\nu$$

An electron can escape from the metal only if a single photon gives it enough energy.

The photoelectric equation is:

$$K_{\max} = h\nu - \phi$$

where:

- $K_{\max}$ is the maximum kinetic energy of the emitted electron,
- $h\nu$ is the photon energy,
- ϕ is the work function of the metal.

If $h\nu < \phi$, the electron does not escape.

That is why low-frequency light does not eject electrons, even if the intensity is high.

Note to Self

Increasing intensity at the correct frequency releases more electrons, because there are more photons. But it does not automatically make each electron more energetic. The energy per photon depends on frequency.

Big Idea

The photoelectric effect showed that light behaves not only like a wave, but also like particles. This became one of the foundations of wave-particle duality.

4 Problem 3: The Atom Problem

The third major failure is what I like to call the atom problem.

This has two parts:

1. Why are atoms stable?
2. Why do atoms emit and absorb only specific wavelengths?

Both are directly important for quantum chemistry.

5 Atomic Stability

5.1 The classical picture

In the early model of the atom, electrons were imagined as tiny particles orbiting the nucleus, almost like planets orbiting the Sun.

But this creates a serious problem.

According to classical electromagnetism, an accelerating charge emits radiation.

An electron moving in a circle is accelerating because its direction keeps changing.

So the electron should continuously lose energy.

If it loses energy, it should spiral inward and crash into the nucleus.

That means classical physics predicts:

Atoms should collapse.

But atoms clearly do not collapse. Matter exists. Chemistry exists. We exist.

So classical physics could not even explain why stable atoms exist.

5.2 How quantum mechanics solved it

Quantum mechanics replaced the idea of fixed electron orbits with **wavefunctions** and **allowed energy states**.

The electron is not treated like a tiny planet moving in a definite orbit.

Instead, it is described by a wavefunction, which gives a probability distribution around the nucleus.

In an atom, the electron can occupy only certain allowed energy states. These are called **stationary states**.

For hydrogen, the energy levels are:

$$E_n = -\frac{13.6 \text{ eV}}{n^2}$$

where:

$$n = 1, 2, 3, \dots$$

The lowest energy state is the ground state.

Once the electron is in the ground state, there is no lower energy state available for it to fall into.

So it cannot keep losing energy continuously and spiral into the nucleus.

5.3 The uncertainty principle view

There is also a deeper way to think about this.

If the electron were squeezed into the nucleus, its position would become extremely well-defined.

That means Δx becomes very small.

But the uncertainty principle says:

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

So if Δx becomes very small, Δp becomes very large.

Large momentum uncertainty means large kinetic energy.

So forcing the electron too close to the nucleus makes its kinetic energy shoot up.

Big Idea

Atomic stability is not a classical orbit problem. It is a quantum balance between attraction to the nucleus and the kinetic energy cost of confining the electron too much.

6 Line Spectra: Starting from Sunlight

6.1 White light is not the full story

Sunlight looks white to us.

But when sunlight passes through a prism, it spreads into a rainbow. At first, this looks like the Sun is giving out every visible colour smoothly.

But when scientists looked carefully, they found something strange.

The solar spectrum has dark lines in it.

These are called **Fraunhofer lines**.

So the question becomes:

Question

Why are some specific colours missing from sunlight?

6.2 Why this was a problem

Classical physics had no natural explanation for this.

If electrons could have any energy, then atoms should absorb and emit light in a continuous way.

But real atoms are selective.

They absorb and emit only specific wavelengths.

It is almost like every element has its own barcode.

Hydrogen has one pattern. Sodium has another. Calcium has another.

6.3 The quantum explanation

Quantum mechanics says that electrons in atoms can have only certain allowed energy levels.

A photon is absorbed only when its energy exactly matches the gap between two energy levels.

The condition is:

$$h\nu = E_2 - E_1$$

If the photon has the wrong energy, the atom does not absorb it.

If the photon has the exact required energy, the electron jumps to a higher energy level.

This explains why atoms absorb only specific wavelengths.

6.4 Why sunlight has dark lines

The inner hot region of the Sun emits an almost continuous spectrum.

As this light passes through the cooler gases in the Sun's outer atmosphere, atoms in those gases absorb very specific wavelengths.

Those wavelengths are removed from the sunlight reaching us.

That is why dark lines appear in the solar spectrum.

Big Idea

The dark lines in sunlight are not random gaps. They are fingerprints of quantized atoms.

6.5 Emission lines are the reverse

If an excited electron falls from a higher energy level to a lower one, it emits a photon.

The photon energy is:

$$h\nu = E_i - E_f$$

Because only certain energy levels exist, only certain photon energies can be emitted.

So we get sharp emission lines.

In short:

Absorption spectrum = missing wavelengths
Emission spectrum = bright wavelengths

Both come from the same idea:

Atomic energy levels are quantized.

7 Quick Summary

Problem	Classical expectation	Quantum idea
Blackbody radiation	Energy should increase without limit at high frequencies	Energy is exchanged in packets, $E = h\nu$
Photoelectric effect	Brighter light should eject electrons regardless of frequency	Light comes as photons, and each photon has energy $h\nu$
Atomic stability	Electrons should radiate energy and collapse into the nucleus	Electrons occupy stationary states, with a ground state below which they cannot fall
Line spectra	Atoms should absorb and emit continuously	Electrons jump between quantized energy levels

Table 1: The main failures of classical physics that pushed physics toward quantum mechanics.

8 Why this matters for Quantum Chemistry

Quantum chemistry is basically what happens when we apply quantum mechanics to atoms and molecules.

This is why this topic matters before starting the actual chemistry.

Classical physics could not explain:

- why atoms are stable,
- why atoms have energy levels,
- why atoms absorb and emit specific wavelengths,
- why electrons form orbitals,
- why chemical bonds form,
- why molecules have specific structures.

Quantum mechanics gives the language for all of this.

Big Idea

Chemistry is largely the behavior of electrons. And electrons are quantum objects. So if I want to understand chemistry from the ground up, quantum mechanics is not optional.

9 Final Takeaway

Classical physics worked beautifully for the macroscopic world, but it failed when nature was studied at the atomic scale.

Blackbody radiation showed that energy exchange is quantized.

The photoelectric effect showed that light behaves like particles.

Atomic stability showed that electrons cannot be understood as tiny planets orbiting the nucleus.

Line spectra showed that atoms have discrete energy levels.

So quantum mechanics was not just a new theory. It was forced on us by experiment.

That is the starting point of quantum chemistry.

**To understand molecules, I first need to understand electrons.
To understand electrons, I need quantum mechanics.**

References and Reading Notes

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- University of Waterloo, notes on classical failures and early quantum theory.
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